

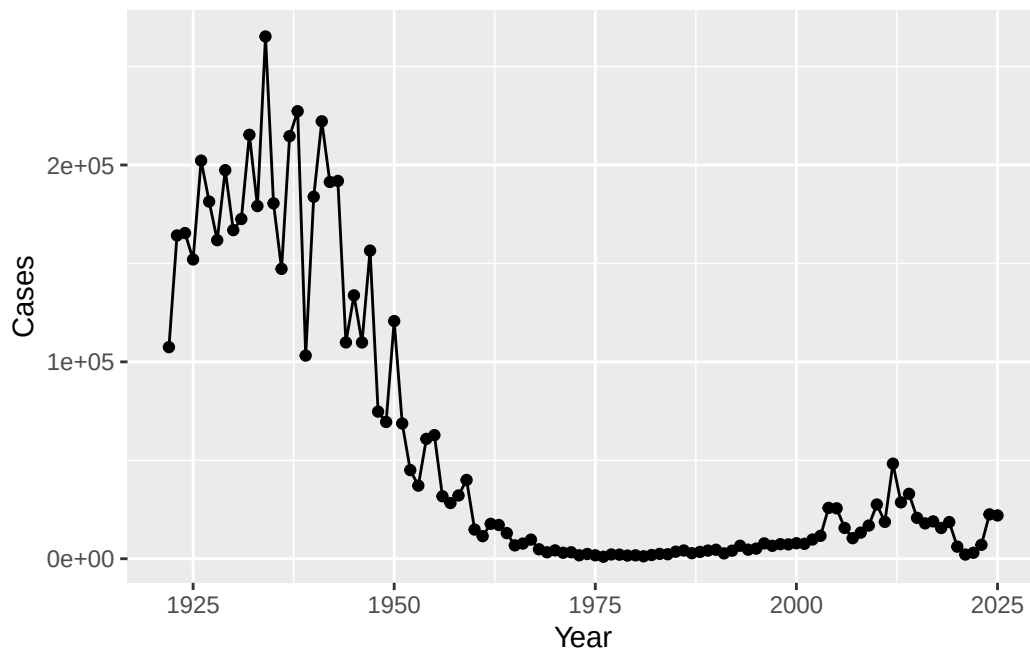
# Class 18: Pertussis Mini-Project

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```
library(ggplot2)
```

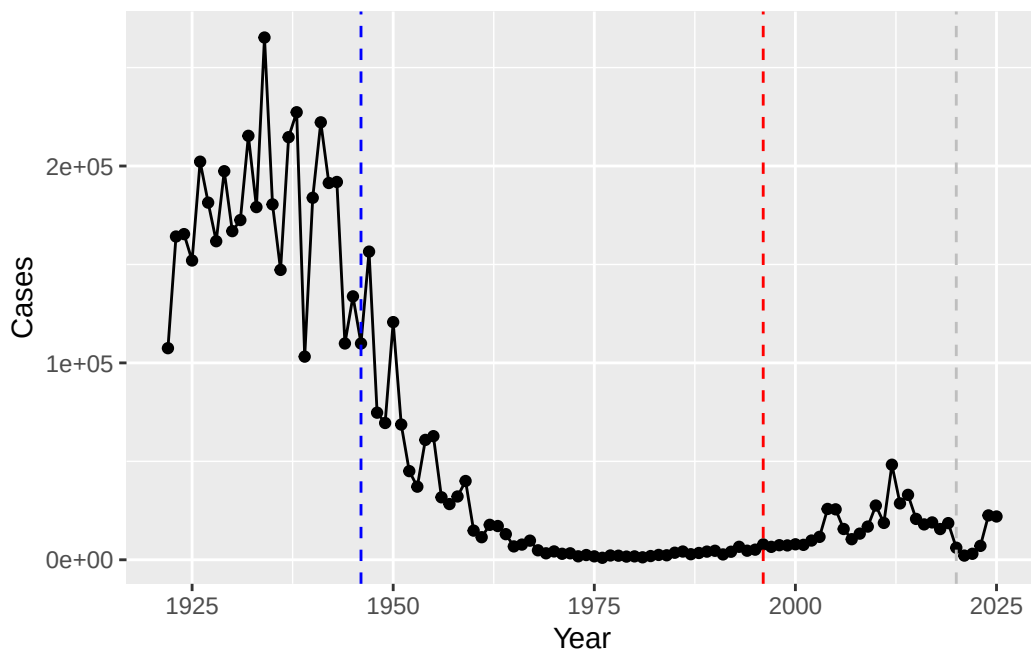
Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
ggplot(cdc) +  
  aes(Year, Cases) +  
  geom_point() +  
  geom_line() +  
  labs(x="Year", y="Cases")
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +  
  aes(Year, Cases) +  
  geom_point() +  
  geom_line() +  
  geom_vline(xintercept = 1946, col = "blue", lty = 2) +  
  geom_vline(xintercept = 1996, col = "red", lty = 2) +  
  geom_vline(xintercept = 2020, col = "gray", lty = 2) +  
  labs(x="Year", y="Cases")
```



After the roll out of the wP vaccine in 1946, cases steadily declined. After the switch to the aP vaccine, the cases could be going up because of better testing, less vaccinations, and vaccine resistance. The aP vaccine also weins off faster than the wP vaccine.

```
library(jsonlite)
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female    Male  
    112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

|                                           | Female | Male |
|-------------------------------------------|--------|------|
| American Indian/Alaska Native             | 0      | 1    |
| Asian                                     | 32     | 12   |
| Black or African American                 | 2      | 3    |
| More Than One Race                        | 15     | 4    |
| Native Hawaiian or Other Pacific Islander | 1      | 1    |
| Unknown or Not Reported                   | 14     | 7    |
| White                                     | 48     | 32   |

```
# Complete the API URLs...  
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)  
titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
abdata <- inner_join(titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

| IgE  | IgG  | IgG1  | IgG2  | IgG3  | IgG4  |
|------|------|-------|-------|-------|-------|
| 6698 | 7265 | 11993 | 12000 | 12000 | 12000 |

Q. How many different “antigen” values are measured?

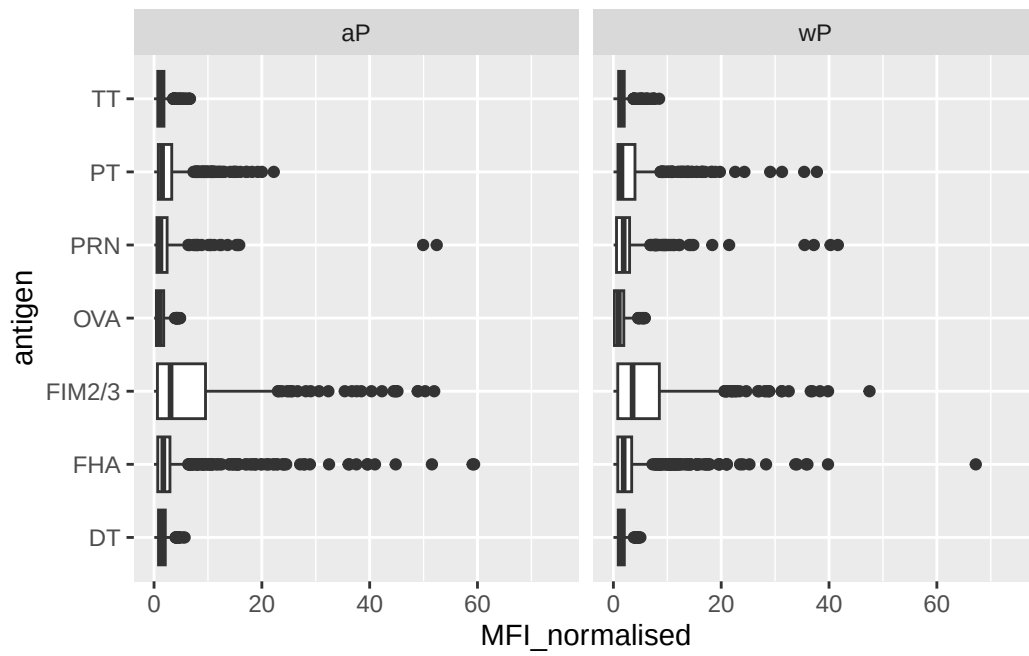
```
table(abdata$antigen)
```

| ACT  | BETV1 | DT   | FELD1 | FHA   | FIM2/3 | LOLP1 | LOS  | Measles | OVA  |
|------|-------|------|-------|-------|--------|-------|------|---------|------|
| 1970 | 1970  | 6318 | 1970  | 6712  | 6318   | 1970  | 1970 | 1970    | 6318 |
| PD1  | PRN   | PT   | PTM   | Total | TT     |       |      |         |      |
| 1970 | 6712  | 6712 | 1970  | 788   | 6318   |       |      |         |      |

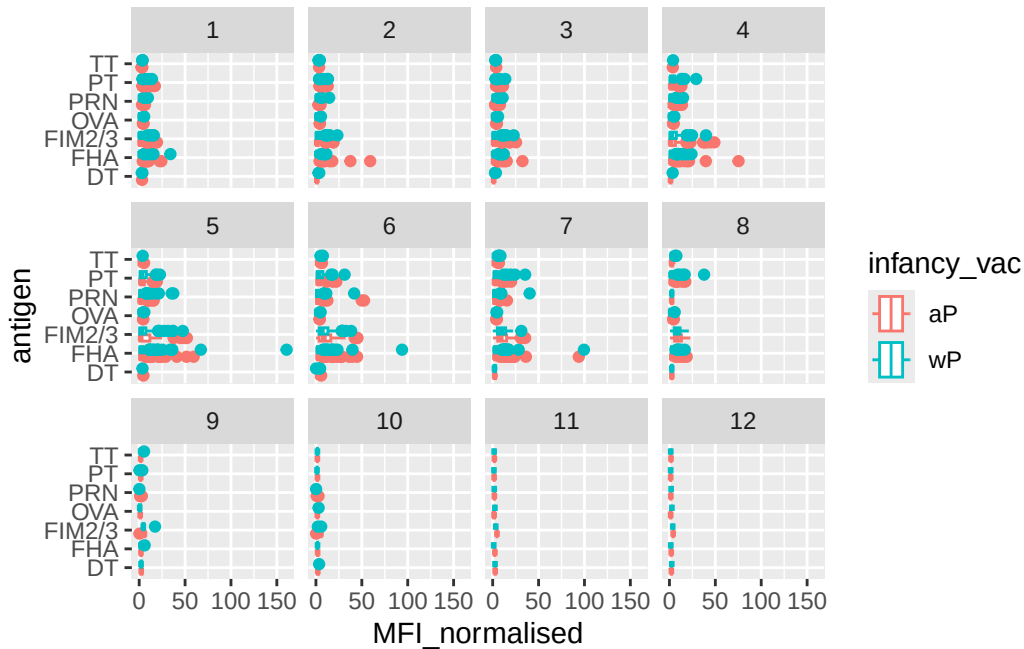
```
igg <- abdata %>% filter(isotype == "IgG")
```

```
ggplot(igg) +  
  aes(MFI_normalised, antigen) +  
  geom_boxplot() +  
  xlim(0,75) +  
  facet_wrap(~infancy_vac)
```

Warning: Removed 5 rows containing non-finite outside the scale range  
(`stat\_boxplot()`).



```
ggplot(igg) +  
  aes(MFI_normalised, antigen, col = infancy_vac) +  
  geom_boxplot() +  
  facet_wrap(~visit)
```



```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    stat_summary(fun = mean, geom = "smooth", linewidth = 2, aes(group = infancy_vac, col = )) +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

## 2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)

